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Problem 1

e. I attempted to further reduce the model size by applying TFLite dynamic range quantization to the knowledge-distilled student model. The student model was already the smallest model from the previous parts because it used a smaller architecture learned from the teacher model.

After quantization, the resulting TFLite model size was 6.14 KB, and the test accuracy was 0.9074. The classification report shows an overall accuracy of 0.91, with macro average F1-score of 0.91 and weighted average F1-score of 0.90.

However, the quantized model did not become smaller than the unquantized student TFLite model. The size reduction was -0.06%, meaning the model became slightly larger. This likely happened because the student model was already very small, so the extra TFLite quantization metadata and operator overhead offset the small amount of weight compression.

Therefore, further size reduction was not meaningfully possible without changing the model architecture more aggressively or risking a larger accuracy drop. The final result still shows that the quantized student model maintained good classification performance, but it did not improve model size beyond the smallest previous model.

Problem 2

1. Under the free-tier Edge Impulse account, several types of learning blocks are available. These include supervised learning blocks such as classification and regression, transfer learning blocks such as image classification and keyword spotting, object detection blocks, anomaly detection blocks, and classical machine learning blocks. For supervised classification, one example is using a Keras classifier to classify accelerometer data into different activities, such as idle, walking, or running. For supervised regression, an example is using a regression model to predict a continuous sensor value such as temperature or vibration level. For transfer learning, an example is using an image classification model to classify objects from camera images, or using a keyword spotting model to recognize spoken words such as “yes” and “no.” For object detection, Edge Impulse supports models such as FOMO or YOLO-based models that can detect objects in an image. For unsupervised or anomaly detection, examples include K-means anomaly detection, Gaussian Mixture Model anomaly detection, and visual anomaly detection, which can be used to identify abnormal sensor readings or visual defects. Classical machine learning can also be used for simpler feature-based classification problems.

2. In the “Classifier” section of Edge Impulse, different neural network layers can be used to build a model. The input layer is used to define the shape of the input features that are passed into the model. Dense, or fully connected, layers connect every neuron from the previous layer to the next layer and are commonly used for classification tasks. Reshape layers are used to change the shape of the data without changing the data values, which is useful when preparing data for convolutional layers. Flatten layers convert multi-dimensional data into a one-dimensional vector before sending it into dense layers. Dropout layers randomly disable some neurons during training, which helps reduce overfitting and

improves generalization. One-dimensional convolution layers are used to extract local patterns from sequential data such as audio, vibration, or other time-series sensor data. One-dimensional pooling layers reduce the size of 1D feature maps and help lower computation. Two-dimensional convolution layers are used to extract spatial features from images or spectrograms. Two-dimensional pooling layers reduce the spatial size of 2D feature maps, which lowers computation and helps make the model more robust. The output layer produces the final prediction, such as the probability of each class in a classification problem.

3. Under the “Deployment” section in the free-tier Edge Impulse account, the main model compression and optimization options include INT8 quantization, Float32 deployment, and EON Compiler optimization. INT8 quantization converts the model from 32-bit floating-point values to 8-bit integer values, which can reduce model size, memory usage, and inference latency. The Float32 option keeps the model in full floating-point precision, which may preserve accuracy but usually requires more memory and computation. Edge Impulse also provides the EON Compiler, which optimizes the model for edge devices and can reduce memory usage during deployment. In some cases, an EON Compiler RAM-optimized mode is also available to further reduce RAM usage. Therefore, the main available methods are INT8 quantization, Float32 deployment, EON Compiler optimization, and EON Compiler RAM optimization.

4. In the free-tier version of Edge Impulse, synthetic data generation mainly supports image and audio input modalities. For image data, Edge Impulse provides synthetic image generation methods such as DALL-E image generation and FLUX Pro image generation, which can create image samples from text prompts. For audio or keyword spotting data,

Edge Impulse provides a Whisper-based keyword spotting generation block, which can generate speech samples for training audio classification or keyword recognition models. Edge Impulse also has time-series data augmentation, but this feature is generally listed as an Enterprise-only option, so it may not be available in the standard free-tier account. Synthetic data is important in machine learning because collecting real-world data can be expensive, slow, or difficult. It can help increase the size of a dataset, balance classes, simulate rare cases, and improve model robustness. For TinyML applications, synthetic data is especially useful because it can help train models when collecting enough real sensor, image, or audio data from the target device is challenging.